Generating images from a subset of the ImageNet Dataset using Deep Convolutional Generative Adversarial Networks (DCGANs)
Assigment 2

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Abstract

Generative Adversarial Networks (GANs) have demonstrated remarkable capacity for high-fidelity image synthesis, yet training stability and sample diversity remain persistent challenges when scaling to large, semantically heterogeneous datasets.
In this work, we investigate the use of conditional GANs for image generation on a curated subset of the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) dataset [1], comprising 100 classes selected to span a range of visual complexity and intra-class variance.
Our training pipeline incorporates a two-timescale update rule (TTUR) to promote generator–discriminator equilibrium.
Qualitative analysis further confirms strong semantic coherence and inter-class disentanglement across generated samples. These findings suggest that targeted subset training is a viable and efficient strategy for domain-specific generative modeling, with implications for data augmentation, few-shot learning, and controllable content generation.

1 Introduction

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. (2014) [2] have become one of the most widely studied frameworks for image generation. By putting a generator network against a discriminator network in a minmax game, GANs are able to learn rich data representations and producing visually convincing synthetic images.

Despite the proliferation of incresingly complex GAN architectures in recent literature, the core framework remains a powerful and instructive baseline for image synthesis tasks.

In this work we apply a standard deep convolutional GAN (DCGAN) [3] to a small subset of the ImageNet dataset. A sample of this data can be seen on fig 1 ImageNet is a large-scale visual database containing over fourteen million labeled images spanning thousands of different categories, and the dataset has also served as a foundational benchmark for image classification, generation and general computer vision research.

Our model employs transposed convolutional layers in the generator to progressively upsample latent vectors into full resolution image, while the discriminator uses strided convolutions to learn and distinguish real samples from generated samples from the generator. Our goal is

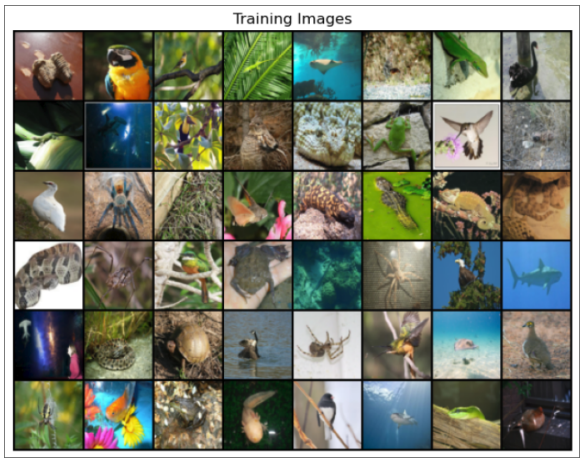

Figure 1: Sample images of the used subsets of the ImageNet dataset

evaluating how well conceptually a simple GAN architecture can capture the appearance of the structure of real-world categories.

2 Literature Survey

2.1 Preceding Generative models

GANs emerged in a moment where there were already pre-existing generative models, each with some significant limitations:

Restricted Boltzmann Machines (RBMs) and **Deep Boltzmann Machines** (DBMs) required Markov chan Monte Carlo sampling, making the generation of synthetic data slow and appproximate.

Variational Autoencoders (VAEs) [4] offered an alternative deep generative framework with principled latent variable inference but often produced blurry samples due to pixel-level reconstruction losses.

Generative Stochastic Networks (GSNs) [5] extended denoising autoencoders but relied on complex approximate inference procedures

The GAN framework offered a compelling alternative by sidestepping explicit likelihood estimation, and it proposed a trainable method that could produce sharper, more realistic samples.

2.2 The original GAN model

The concept of Generative Adversarial Networks was introduced by Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville and Yoshoua Beingo in 2014. The foundational paper titled "Generative Adversarial Nets" [2] proposed a framework for training generative models through an adversarial process involving two models that competed between each other in a minimax game.

The two models of the GAN are trained simultaneously, each one having a different purpose:

- A generative model G captures the underlying data distribution of the training set. Its goal is to transform a noise vector z, called latent space, into synthetic samples.
- A discriminative model D that estimates the probability that a given sample is coming from real data or if it is synthetic data generated from the generator G.

The adversarial objective can be formally expressed as:

$$\min_G \max_D V(D, G) =$$
$$= \mathbb{E}(\log D(x)) + \mathbb{E}(\log(1 - D(G(z))))$$

This adversarial min max game converges when the generator perfectly recovers the training data distribution and the discriminator assigns a probability of 1/2 to each given sample, so it is unable to discriminate between real and synthetic data.

The original implementation of GAN networks used Multilayer Perceptrons (MLPs) as both the generator and the discriminator. Experiments were conducted on three datasets:

- The MNIST [6] dataset, which contains labeled images of handwritten digits
- The Toronto Face Dataset (TFD) [7]
- The CIFAR-10 natural image dataset [8]

The results of the original paper were promising but limited on image quality, particularly for complex natural image domains. It demonstrated the potential of the framework while highlighting the need for architectural improvements to scale to richer visual content.

2.3 Early GAN variants and successors

Conditional GAN (cGAN): [9] Conditional GANs extend the original GAN framework by conditioning both the generator and the discriminator on auxiliar information such as class labels. This allowed directed generation of specific classes of data, converting the unsupervised formulation into one capable of target synthesis. cGANs demonstrated that class dependant image generation could be achieved with a simple architectural extension.

Laplacian Pyramid GANs (LAPGAN): [10] LAPGAN employed a cascade of convolutional networks with a Laplacian pyramid framework, generating images in a coarse-to-fine fashion. While LAPGAN achieved visually compelling results, the multi-network approach added more complexity. This framework demonstrated that convolutional architectures were promising for adversarial image generation, directly motivating the development of an unified deep convolutional architecture, that later would become DCGANs.

2.4 Deep Convolutional Generative Adversarial Netwokrs (DCGAN)

Despite the theoretical elegance of the original GAN, practical training proved to be difficult. The use of fully connected architectures imposed strong independence assumptions, inappropiate for image data. The input data of these models needed to have limited resolution because of its dense structure. The original GAN models also produced noisy and incoherent outputs over complex training datasets.

The integration of CNNs, which had demonstrated unparalleled feature learning for images in supervised settings (eg. AlexNet, VGG or GoogLeNet), into the GAN framework was a natural next step but required some careful architectural design.

DCGANs were first addressed in the paper "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks". This paper has since accumulated over 14,000 citations, cementing its status as a landmark contribution in deep learning.

The central contribution of DCGAN is a set of empirically validated architectural guidelines that stabilize GAN training when using deep convolutional networks:

- Replace all pooling layers with strided convolutions in the discriminator and fractional-strided (transposed) convolutions in the generator model. This allows both networks to learn their own spatial downsampling and upsampling rather than relying on fixed pooling operations. Strided convolutions preserves more spatial information and allows learning spatial hirearchies.

- Use batch normalization in both the generator and the discriminator. Batch normalization stabilizes training by normalizing the input to each unit, mitigating issues with poor weight initialization and allowing gradients to flow more reliably through deeper networks.

- Remove fully connected hidden layers for deeper architectures. The original GAN models used dense layers to connect the latent vector to the convolutional layers. DCGANs instead connects directly from the latent vector to the first convolutional layer using a reshape operation. This enables a fully convolutional pipeline.

- Use ReLU activation in the generator for all layers except the output, which uses tanh. Use LeakyReLu (with a negative slope of 0.2) in the discriminator for all layers, with a Sigmoid output for binary classification of real and fake data.

2.5 Datasets in GAN and DCGAN Research

The choice of picking a dataset is fundamental to GAN research, as it determines the complexity of the distribution function to be learned, the resolution of the data and diversity of the input samples, and the applicability of evaluation metrics. Different GAN architectures have been evaluated on progressively more challenging datasets as the field has advanced.

MNIST and CIFAR-10. [6] [8] The MNIST dataset, comprising of 70000 grayscale images of handwritten digits at 28×28 resolution was one of the first datasets used to evaluate the original GAN model in 2014. Its simplicity made it an ideal proof-of-concept benchmark. On the other hand, the CIFAR-10 dataset consisted of 60000 color images across 10 object classes at 32×32 resoluton. This dataset provided a more challenging benchmark. Both datasets are continued to serve as standard evaluation platforms in deep learning research due to their reference metrics and also wide adoption.

LSUN - Large-Scale Scene Understanding [11] The LSUN dataset was one of the three primary datasets used to evaluate DCGAN in the original paper. The bedroom susbet contains aproximately 3 million annotated scene images, providing a large-scale training corpus. DCGAN trained on LSUN bedrooms was among the first GAN models to generate photorealistic interior scenes at a 64×64 resolution, demonstrating the capability of the convolutional adversarial approach.

ImageNet-1k [1] ImageNet-1k contains approximately 1.2 million images across 1000 object categories. It was used in the DCGAN paper to evaluate the quality of learned representations for supervised tasks. The discriminator's convolutional features were used to train a regularized linear $L2 - SVM$ classifier on the CIFAR-10 datset. This classifier achieved an 82% accuracy without any fine-tunning, demonstrating the transferability of representations learned using adversarial training.

Toronto Face Dataset (TFD) [7] The Toronto Face Dataset was used in the original GAN paper as an early benchmark for face generation. TFD contains approximately 100000 grayscale facial images representing various expressions and identitites. While it served as a foundational role in early GAN evaluations, it has been replaced by larger and more diverse datasets such as CelebA and FFHQ for face generation research.

3 Methodology

We tested the DCGAN novel architecture against 100 random classes of the ImageNet dataset. Our objective was to make the generator learn the distribution of these 100 classes, so we can generate synthetic data taken from these subset. The implementation was made using the PyTorch python library. Additional libraries such as NumPy handled numerical operations, and Matplotlib handled data visualization.

3.1 Parameters

Each layer of the generator has a convolutional transpose, followed by a batch normalization and a ReLU activation function. It has 5 of these layers:

- ConvTranspose2d($100 \to 512, k = 4, s = 1, p = 0$)
- ConvTranspose2d($512 \to 256, k = 4, s = 2, p = 1$)
- ConvTranspose2d($256 \to 128, k = 4, s = 2, p = 1$)
- ConvTranspose2d($128 \to 64, k = 4, s = 2, p = 1$)
- ConvTranspose2d($64 \to 3, k = 4, s = 2, p = 1$) (Tanh no batchnorm)
- **Output:** $3 \times 64 \times 64$

And the discriminator takes as an input the generated image, does some regular convolutions with batch normalization and a LeakyReLU with a negative slope of 0.2, and finally a sigmoid activation function:

- Conv2d($3 \to 64, k = 4, s = 2, p = 1$) (No batchnorm)
- Conv2d($64 \to 128, k = 4, s = 2, p = 1$)

- Conv2d($128 \to 256, k = 4, s = 2, p = 1$)
- Conv2d($256 \to 512, k = 4, s = 2, p = 1$)
- Conv2d($512 \to 1, k = 4, s = 2, p = 0$)
- **Output:** 1

4 Experimental results

We measured the loss of the generator and the discriminator during each epoch of training. The results can be seen on fig. 2 and 3.

An evolution of the model can be seen on the appendix on fig. 4

5 Conclusions

As we can see, DCGANs provide a rich framework model for creating generative models of synthetic data that capture really well the underlying features of the training data, and also can achieve good results on approximating the distribution function of the input dataset, as we have seen by training a test DCGAN model over a small slice of the ImageNet dataset.

The implementation of the project (and the source of this document) can be found on `https://github.com/BinarySandia04/Assigment2ALLM`

Figure 2: The generator loss during training.

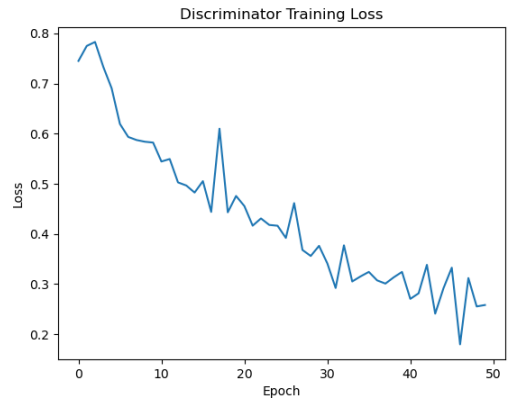

Figure 3: The discriminator loss during training.

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A Appendix

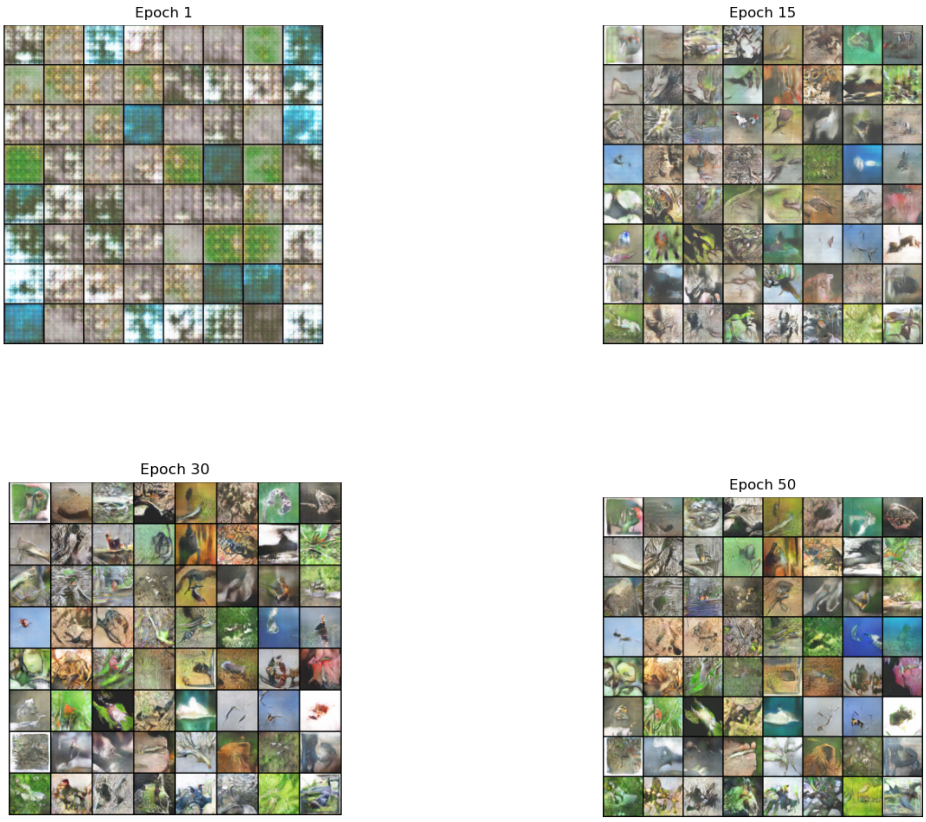

Figure 4: Evolution of the output of the generator over the same noise sample on different epochs. One can see that on later epochs the quality of the synthetic images improves.